Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/08/2025
PI Name:	Strandwitz, Philip Ph.D.		
Project Title:	Development of Next-Generation Probiotics to Combat Mental Health Risks for Space Travel		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	
Organization Name:	Holobiome, Inc.		
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PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02139	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Industry
Start Date:	11/01/2020	**End Date:**	10/31/2021
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gilbert, Jack Ph.D. (University of California, San Diego)		
Grant/Contract No.:	NNX16AO69A-IND0103		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Industry Project Holobiome is developing next generation probiotics, derived from the healthy human microbiome, to address major global health problems. With this application we seek to explore utility of these strains to solve a major challenge for the intergalactic advancement of humanity -- the mental health of deep space explorers. We call these Astrobiotics.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	The goal of this project was to support development of the first "Astrobiotic", designed to combat and/or treat cognitive or behavioral conditions and psychiatric disorders astronauts may experience during their journeys. Resources acquired via this solicitation were used to perform lead optimization on HOLO-101 -- a consortium of two human-derived strains which target the host GABAergic and inflammatory response -- in two rat models of stress and depression. The goals were twofold: (1) determine whether HOLO-1 could exhibit efficacy in two animal models of behavior, as compared to the positive control ketamine; and (2) if so, determine whether a single strain representative within HOLO-1 could exhibit the same effect. Two animal models (rats) were used in this study. The first was the limited bedding and nesting model, which elicits early life stress and is well published to induce depressive- and anxiety-like behavior. Unfortunately, after significant effort in optimizing the model using published protocols and consulting with experts, the model was not seen to cause depressive- or anxiety-like behavior. We then moved on to a second model, known as social defeat, which we did find exhibited the anticipated phenotype. Excitingly, in this model we found HOLO-1 exhibited anti-depressant activity, as measured via the sucrose preference test. Analysis is ongoing for a range of tissues collected throughout the study, including shotgun metagenomic sequencing of cecal contents and stool; serum, cecum, stool, and brain GABA levels; and RNAseq in cecum and brain tissues. Immunological profiling is ongoing in plasma samples. This analysis is expected to be completed in Q1 2021. We anticipate the intervention with HOLO-101 will be associated with increased systemic levels of GABA and/or increase GABAergic activity in some of the collected tissues. Supported by additional Holobiome studies showing therapeutic potential of HOLO-1, these results encourage us to enter into manufacturing and human testing in 2022. Given the path-to-market is currently first planned as a consumer facing probiotic, and assuming strong safety profiles in humans, HOLO-1 may be available to the public in 2023.
Bibliography Type:	Description: (Last Updated: 03/25/2026)
Articles in Peer-reviewed Journals	Pathare NN, Fayet-Moore F, Fogarty JA, Jacka FN, Strandwitz P, Strangman GE, Donoviel DB. "Nourishing the brain on deep space missions: Nutritional psychiatry in promoting resilience." Front Neural Circuits. 2023 Aug 17;17:1170395. https://doi.org/10.3389/fncir.2023.1170395 ; PMID: 37663891; PMCID: PMC10469890 , Aug-2023
Articles in Peer-reviewed Journals	Dothard MI, Caboni M, Norment D, Sigmund N, Allard SM, Gilbert JA, Gavrish E, Al-Ghalith G, Der-Avakian A, Strandwitz P. "A human-derived Bacteroides strain attenuates depressive-like behavior in a rat model of social defeat-induced stress." BMC Medicine. 2026 Mar 10. Online ahead of print. https://doi.org/10.1186/s12916-026-04787-v ; PMID: 41808137 , Mar-2026